

Linux Basic Commands

✓ 1. Navigation Commands

`pwd` – Print Working Directory

Shows the current location in the filesystem.

```
pwd
```

📌 Output example:

```
/Users/yourname/projects
```

`ls` – List Directory Contents

Lists files and folders in the current directory.

```
ls
```

- `ls -l` → Detailed list (permissions, size, date)
 - `ls -a` → Shows hidden files (those starting with `.`)
 - `ls -la` → Combined
-

`cd` – Change Directory

Moves into a directory.

```
cd folder_name
```

Examples:

```
cd Documents      # Go to Documents
cd ..             # Go up one level
cd /              # Go to root
cd ~              # Go to home directory
```

✓ 2. File and Directory Management

mkdir – Make Directory

Creates a new folder.

```
mkdir new_folder
```

touch – Create File

Creates an empty file.

```
touch file.txt
```

cp – Copy Files or Directories

```
cp source.txt destination.txt
```

- Copy folder:

```
cp -r folder1 folder2
```

mv – Move or Rename Files

```
mv oldname.txt newname.txt
```

```
mv file.txt ~/Documents/      # Move file
```

rm – Remove Files

```
rm file.txt          # Delete file  
rm -r folder_name    # Delete folder (recursively)
```

⚠ **Be careful!** There is no undo.

✓ 3. File Viewing & Editing

cat – View File Contents

Displays content in terminal.

```
cat file.txt
```

nano – Edit Files in Terminal

A basic terminal-based text editor.

```
nano file.txt
```

- Use arrows to move
 - **CTRL + O** to save
 - **CTRL + X** to exit
-

clear – Clears the Terminal

```
clear
```

Shortcut: **CTRL + L**

✓ 4. System Commands

echo – Print Text

Useful for debugging or scripting.

```
echo "Hello, World!"
```

```
Last login: Wed Sep 10 14:31:11 on ttys005
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % nano file.txt
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % chmod +x file.txt
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % ./file.txt
Hello, World!
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % █
```

whoami – Show Current User

```
whoami
```

```
Last login: Wed Sep 10 14:31:11 on ttys005
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % nano file.txt
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % chmod +x file.txt
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % ./file.txt
Hello, World!
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % clear
```

```
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % whoami
vaibhavkumar
(vaibhavkumar@VAIBHAVs-MacBook-Air ~ % █
```

man – Manual for Any Command

```
man ls
```

Use **q** to quit the manual.

✓ 5. Searching and Finding

find – Locate Files

```
find . -name "*.txt"
```

🔍 Finds all **.txt** files in current folder and subfolders.

grep – Search Inside Files

```
grep "hello" file.txt
```

🔍 Searches for the word **hello** inside **file.txt**.

✓ 6. Helpful Shortcuts

Shortcut	Action
Tab	Auto-complete files/folders
↑ / ↓	Browse command history
CTRL + C	Stop a running command
CTRL + L	Clear screen

✓ 7. Bonus: Chaining Commands

- Run multiple commands:

```
mkdir test && cd test && touch hello.txt
```

- Run only if previous command succeeds: &&
- Run regardless of success: ;